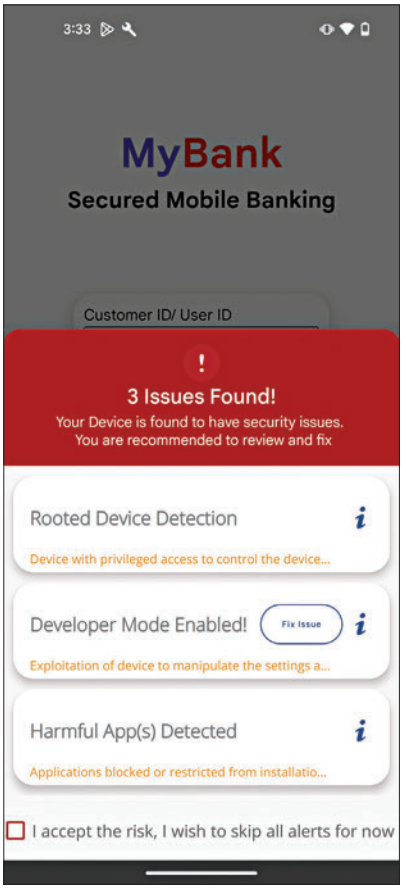
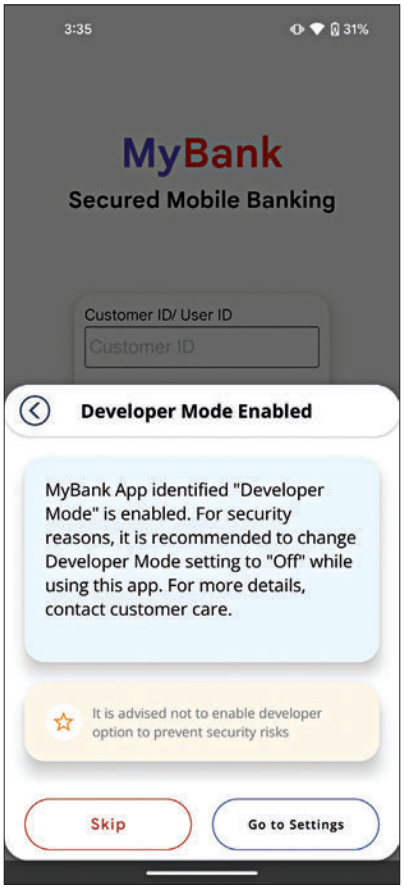




Another major challenge is that these malware variants continuously change their signatures. Once one malware is identified, it can change its name and signature, disappear and reappear. Three years ago, changing signatures was a relatively difficult task. In the AI era, this can happen within a couple of hours.

The second major challenge is the significant increase in social engineering and phishing.

People receive scam and fraudulent calls impersonating banks, government departments, government services, police departments or agencies such as the CBI. Consumers can be socially engineered into sharing sensitive banking or personal information, which is then exploited.

These are the two key challenges we are seeing: malware being circulated to target devices and applications, and social engineering attacks.

Both are also related to privacy and awareness. People sometimes click on links, download APKs, install applications and grant permissions without fully understanding the risks. Lack of awareness is another significant challenge.

Do these attacks target specific demographics, or is it "spray and pray"?

Manish: Trojans and malicious URLs are circulated in the thousands. Whoever clicks on them and installs the malware becomes a potential target. For attackers, even a 1% success rate can represent a significant success because these links are circulated through emails, SMS and WhatsApp.

In the case of social engineering, we have observed some demographic patterns. People aged 55–60 years and above are often targeted and can sometimes become easier prey.

We have also seen cases in Tier 3 and Tier 4 cities, largely involving smaller transaction amounts. In many instances, victims receive calls impersonating government schemes or bank customer service representatives. They install malicious applications because they believe they are receiving a genuine call from a bank or government service.

Overall, given India's large population and the scale at which these links are distributed, malware attacks tend to be more random. Whoever gets caught becomes the target.

Fake shopping sites pop up before festive sales and vanish. Are bad actors increasingly exploiting discount deals?

Manish: The festive season and discounting season create significant opportunities for bad actors because consumers are actively shopping.

Fraudsters are building duplicate and malicious websites, and with AI, they can create hundreds of websites within a couple of days and host them outside the country. They are also creating downloadable APKs. Consumers may be asked to provide their information, download an application and receive a ₹100 discount code in return. Consumers can also be attracted by heavy discounts, such as 50% or 70% off, or ₹1,000 or ₹2,000 shopping coupons.

Fraudsters also use schemes where consumers are asked to provide the mobile numbers of five friends or family members in exchange for a ₹100 coupon. People then start sharing this information.

I would describe this as a farming or seed-growing season for fraudsters. From the beginning of the shopping season through December, we see a significant amount of activity. This includes fake websites, fake calls, fake mobile applications, fraudulent schemes and data collection.

Fraudsters may ask for card details, Aadhaar details and other personal information in exchange for attractive discounts or cash rewards. The idea is largely psychological: convince consumers that they are receiving money,